

DATA STRUCTURES

Module 2

Introduction to Linked List

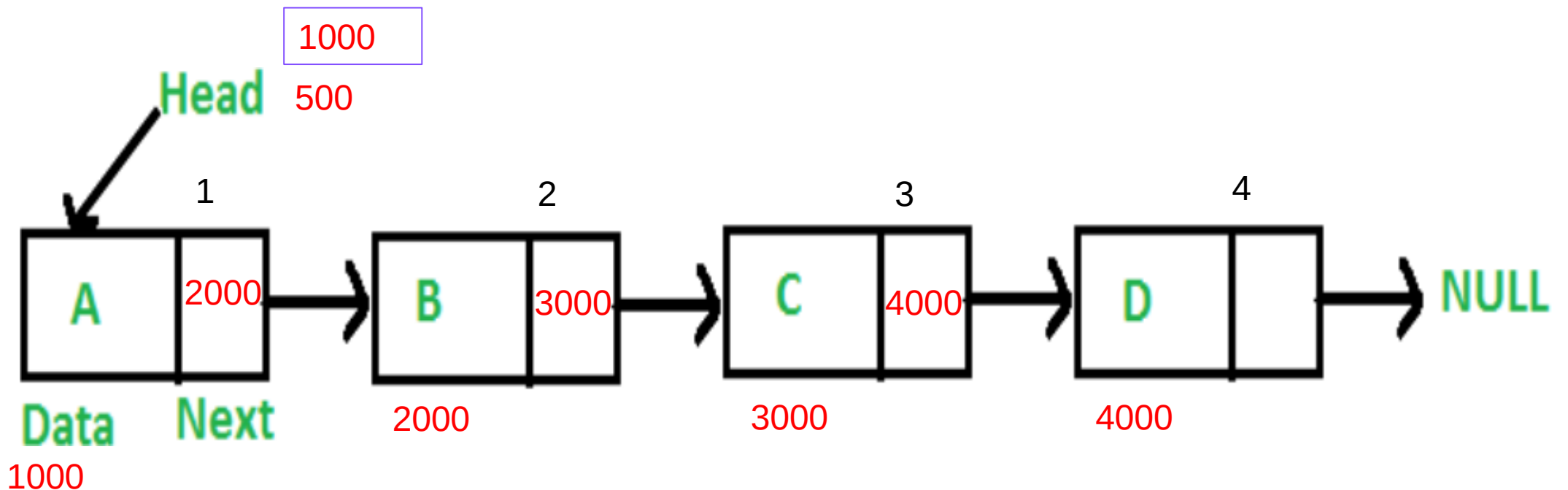


Need of data structures

- To better organize the data
- Data structures: Linear data structures Vs Non Linear Data Structures

Linked Lists

- Unlike arrays, linked list elements are not stored at contiguous location, the elements are linked using pointers



Arrays Vs Linked List

Arrays	Linked list
Fixed size: Resizing is expensive	Dynamic size
Insertions and Deletions are inefficient: Elements are usually shifted	Insertions and Deletions are efficient: No shifting
Random access i.e., efficient indexing	No random access → Not suitable for operations requiring accessing elements by index such as sorting
No memory waste if the array is full or almost full; otherwise may result in much memory waste.	Since memory is allocated dynamically(acc. to our need) there is no waste of memory.
Sequential access is faster [Reason: Elements in contiguous memory locations]	Sequential access is slow [Reason: Elements not in contiguous memory locations]

Source: GeekforGeeks